Time Series Agent: Components and Information Flow

scope of reuse: general → temporal → task-specific



Time series & context

observations · events · task request

LLM Side



§3

interprets temporal evidence and selects actions

- Representation alignment
- Temporal instruction & reasoning
- Workflow policy learning

ChatTS · Time-LLM · TimeRLM

General Harness



§4

executes actions, keeps state, coordinates roles

- Interface & Interaction
- Memory & Context
- Tools & Execution
- Planning & Orchestration
- Optimization & Evolution

TimeClaw-Harness · AION · TS-Agent · TradingAgents

General Time Series Harness



§5

supplies temporal data, tools and time constraints

- Temporal Data Management
- Temporal Processing
- Temporal Context & Memory
- Temporal Validation

TimeSage-EV · OpenForecaster · TSQAgent · TS-Reasoner

Task-Specific Harness



§6

organizes outputs around task objectives

- Forecasting & Prediction
- Data Augmentation & Synthesis
- Anomaly Detection & Diagnosis
- Decision Support

FLAIRR-TS · MetaCaster · Argos · FinCon

Validation & Evaluation



§7

checks process, temporal validity, task and decision utility

- Execution checks
- Information-cutoff checks
- Task metrics
- Downstream utility

TimeSeriesGym · Dr-CiK · TraceBench · TSFLab



Forecast · diagnosis · decision

accepted output



act, observe, check, revise

↺ feedback from validation: revise the plan, or terminate

Solid: execution flow · dashed: returned observation · purple loop: validation feedback